

Sublevel sets of the CATE function

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Motivation

The average treatment effect provides a (too) simple summary of how a treatment works in a population.

Interest in investigating heterogeneity in treatment effects (and finding subpopulations with different treatment effects).

Active research area on estimating the conditional average treatment effect (CATE).

The CATE function is a complicated object, in particular when many baseline covariates are recorded.

Objective to estimate lower-dimensional *summaries* or *features* of the CATE function which can be visualized and interpreted.

Formal setup

$(Y^1, Y^0, W) \sim Q$, Y^a counterfactual binary outcome under treatment levels $a \in \{0, 1\}$, $W \in \mathcal{W} \subset \mathbb{R}^d$.

The CATE function is $w \mapsto \mathbb{E}_Q[Y^1 - Y^0 \mid W = w]$.

We observe $O = (W, A, Y) \sim P$ where $A \in \{0, 1\}$ is the actually given treatment and $Y = Y^1A + Y^0(1 - A)$.

Under standard causal assumption, the function-valued parameter τ defined as

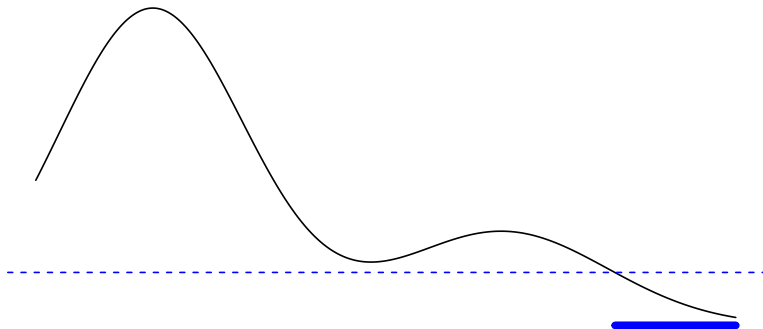
$$\tau(P)(w) = \mathbb{E}_P[Y \mid A = 1, W = w] - \mathbb{E}_P[Y \mid A = 0, W = w]$$

identifies the CATE function.

Idea: Study the area of the sublevel sets of τ

The (sub)level set of the function $\tau(P): \mathcal{W} \rightarrow \mathbb{R}$ for level $\alpha \in [-1, 1]$ is

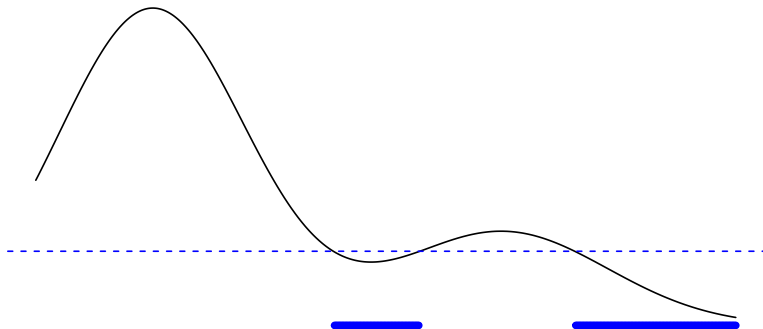
$$\mathcal{W}_\alpha(P) = \{w \in W : \tau(P)(w) \leq \alpha\}.$$



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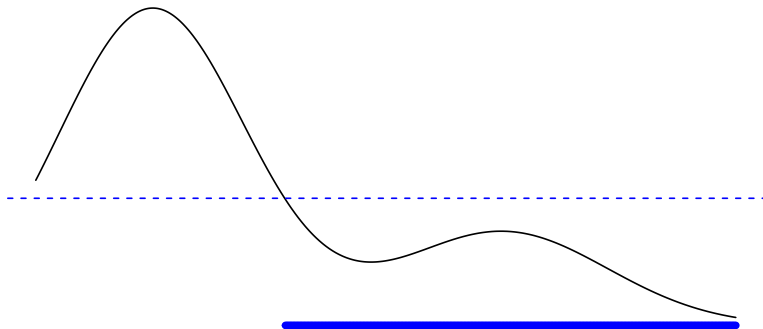
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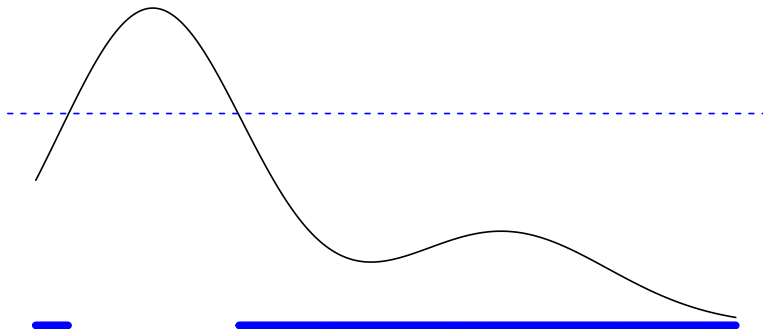
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Estimand

We consider the parameter γ defined as the function $\gamma(P): [-1, 1] \rightarrow [0, 1]$ with

$$\gamma(P)(\alpha) = P(\mathcal{W}_\alpha(P)) = P(\tau(P)(W) \leq \alpha).$$

Alternative formulation is that $\gamma(P)$ is the cumulative distribution function of the random variable $\tau(P)(W) \in [-1, 1]$.

Univariate, nondecreasing function which is easy to visualize.

The value $\gamma(P)(\alpha)$ is the proportion of the population with an expected treatment effect below or equal to α . For instance, $\gamma(P)(0)$ is the proportion of the population for which the treatment is not expected to be beneficial.

Illustration of the function $\gamma(P): [-1, 1] \rightarrow [0, 1]$

`glm(Y ~ age * A + sex)`

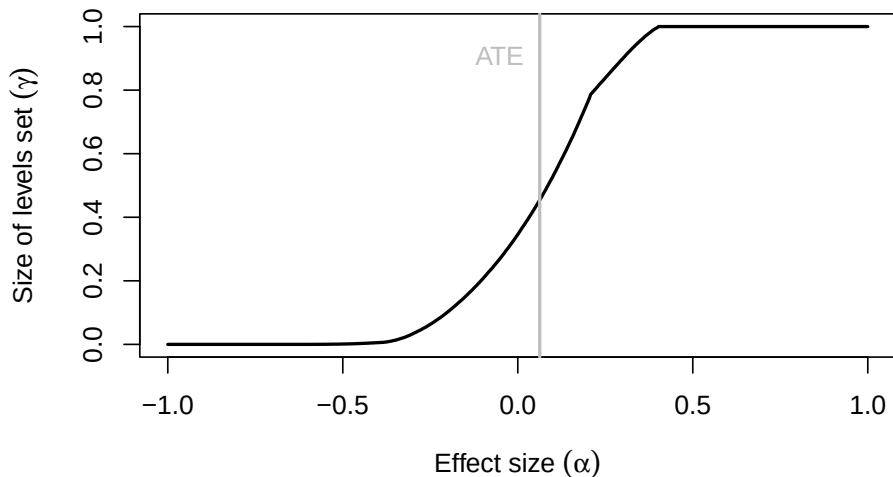


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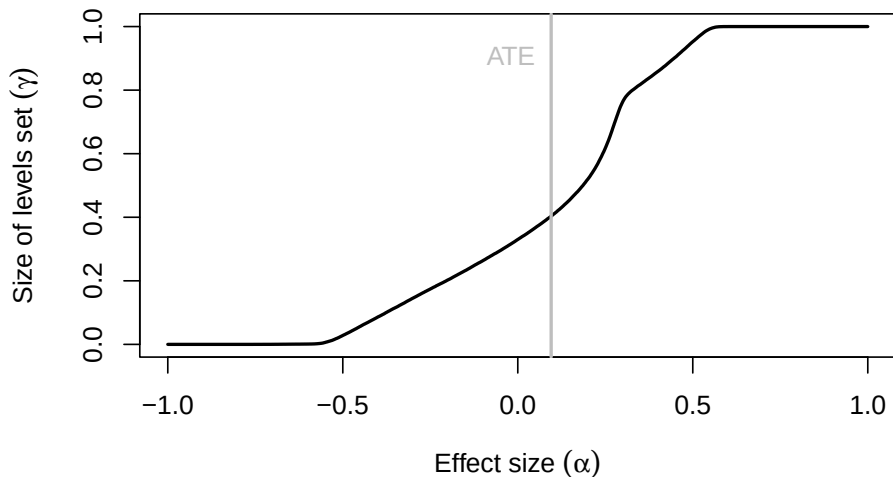


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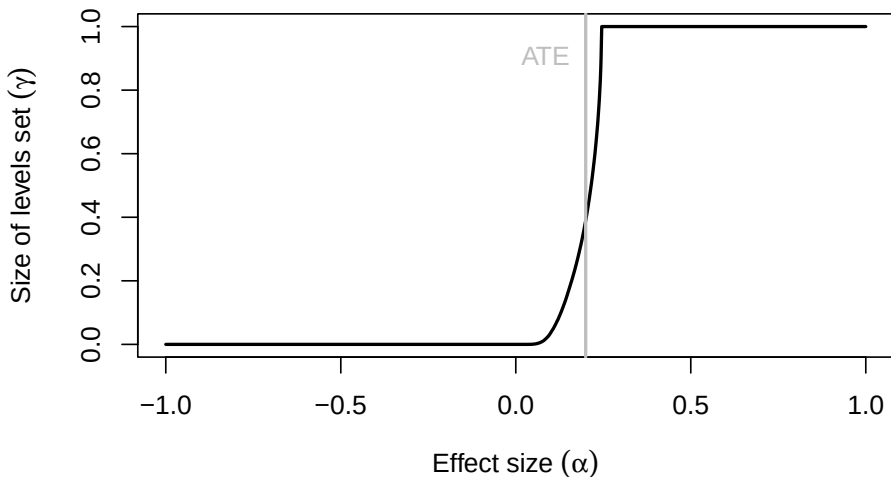
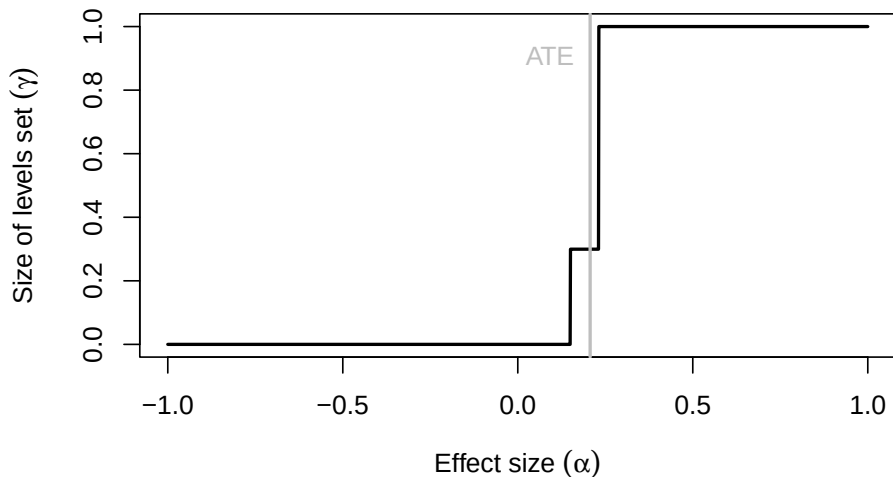


Illustration of the function $\gamma(P): [-1, 1] \rightarrow [0, 1]$

`glm(Y ~ sex + A)`



Estimation

For fixed $\alpha \in (-1, 1)$, the parameter $P \mapsto \gamma(P)(\alpha)$ is not pathwise differentiable (under some regularity conditions*).

Necessary condition for the existence of a regular and asymptotically linear estimator [van der Vaart, 1991].

$\implies$ Usual toolbox not available.

*Led me to the book “Differential Topology” [Guillemin and Pollack, 1974]:

The intent of this book is to provide an elementary and intuitive approach to differential topology. The topics covered are nowadays usually discussed in graduate algebraic topology courses as by-products of the big machinery, the homology and cohomology functors. For example, the Borsuk-Ulam theorem drops out of the multiplicative structure on the cohomology ring of projective space; the Lefschetz theorem comes from Poincaré duality and the Künneth theorem; the Jordan-Brouwer separation theorem follows from Alexander duality; and so on. We have two objections

Estimators

The standard toolbox is not available → many interesting procedures to consider:

- “Naive plug-in” estimator
- Generalized Grenander-type estimator
- Estimate a best approximating spline estimand
- Orthogonal statistical learning

Naive plug-in (i.e., the obvious) estimator

```
glm_fit <- glm(Y ~ A*Age*Sex, data = sim_dd, family = binomial())

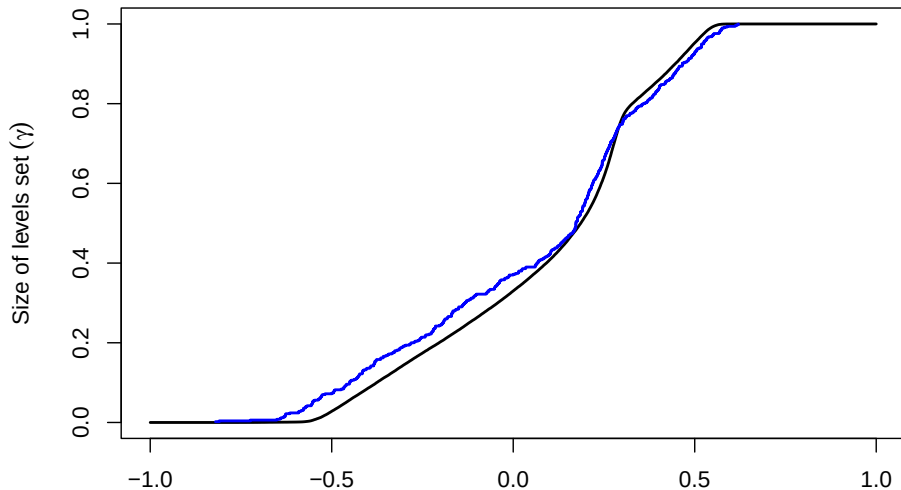
wd0 <- copy(sim_dd)[, A := 0]
wd1 <- copy(sim_dd)[, A := 1]

Y1_hat <- predict(glm_fit, newdata = wd1, type = "response")
Y0_hat <- predict(glm_fit, newdata = wd0, type = "response")
est_cates <- Y1_hat - Y0_hat

Fn <- ecdf(est_cates)
plot(Fn)
```

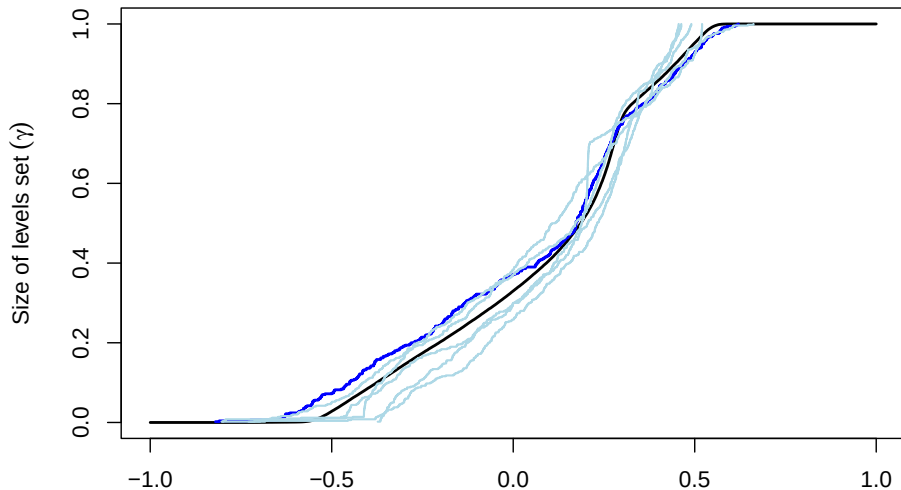
Naive plug-in (i.e., the obvious) estimator

Estimating with correctly specified glm



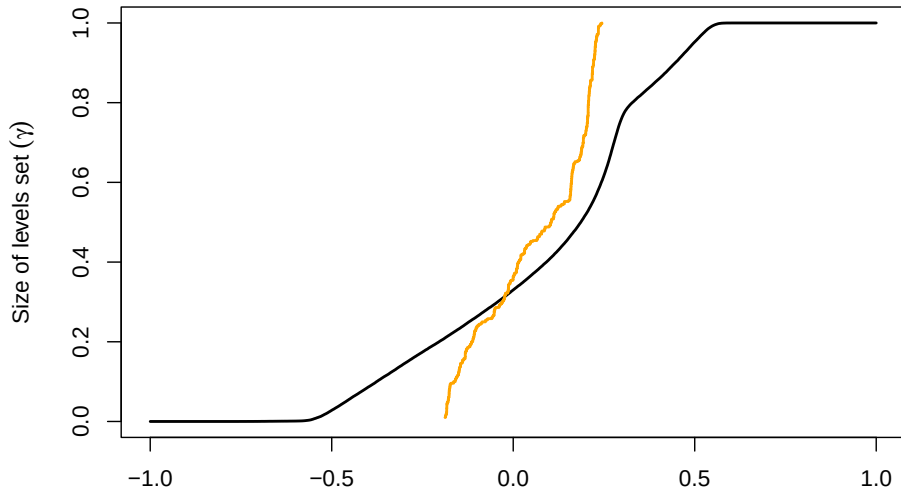
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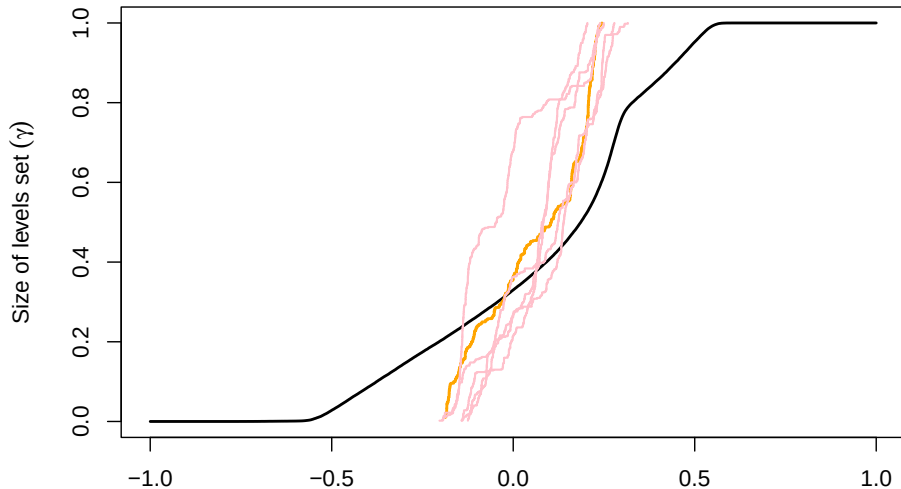
Naive plug-in (i.e., the obvious) estimator

Estimating with random forest



Naive plug-in (i.e., the obvious) estimator

Estimating with random forest



Generalized Grenander-type estimator[†]

The function $\alpha \mapsto \gamma(P)(\alpha)$ is nondecreasing. The primitive of a nondecreasing function is convex.

Idea:

1. Estimate the primitive function (aka anti-derivative) $\alpha \mapsto \Gamma(P)(\alpha) = \int_{-1}^{\alpha} \gamma(P)(u) \, du$
2. Find the greatest convex minorant of the estimator $\hat{\Gamma}_n$
3. Estimate γ with the derivative of $\hat{\Gamma}_n$

The parameter $P \mapsto \Gamma(P)(\alpha)$ is pathwise differentiable at points α at which $\alpha \mapsto \gamma(P)(\alpha)$ is continuous.

We can use the standard toolbox to estimate $\Gamma(P)$.

[†]Westling and Carone [2020], van der Vaart and van der Laan [2006], Groeneboom [1983], Grenander [1956]

Approximate estimand

Estimate an approximation of the function $\gamma(P)$, e.g., the best 0- or 1-order spline approximation to $\gamma(P)$.

Splines (with fixed knots) are determined by a set of coefficients. These coefficients are pathwise differentiable, if the knots are not placed at points of discontinuity of $\alpha \mapsto \gamma(P)(\alpha)$.

Orthogonal statistical learning [Foster and Syrgkanis, 2023]

Extending some of the ideas from the debiased machine learning literature to more general *statistical learning* – estimating function with empirical loss minimization.

Challenges

- Not clear how incorporate the very interesting special case that $\alpha \mapsto \gamma(P)(\alpha)$ is the CDF of a Dirac measure = no heterogeneity.
- Inference

Next project

The function γ can be used to visually investigate if there is any treatment heterogeneity in the population.

But it does not tell us who the people are that respond differently.

Approximate the levels sets $\mathcal{W}_\alpha(P)$ with boxes for suitable choices of α to identify subgroups that can be expected to respond differently to treatment.

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